# OpenVisualJev: Design and Analysis of a Multimodal Decision Model

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## Abstract

Multimodal decision models aim to support low-latency decisions by directly mapping visual context and runtime-defined choices to probabilities. Building such models requires a scoring interface that accommodates changing candidate sets and produces reliable decision scores. We present OpenVisualJev, a multimodal decision model built on a vision–language backbone with a dynamic candidate-scoring head. Tree-masked attention shares image and question prefixes across isolated candidate branches; permutation-equivariant set interaction then compares the resulting representations and preserves consistency under candidate reordering. Calibration-aware training and held-out temperature scaling refine the resulting probabilities. Experiments on visual relation tasks, fine-grained recognition, and natural visual question answering show that this design preserves decisions under candidate reordering, learns set-dependent preferences, and remains competitive with a complete-candidate language-model readout on standard tasks. Its advantage becomes more pronounced with larger difficult candidate sets. Component analyses further clarify the contributions of candidate interaction, variable-count training, and probability calibration. These findings support explicit candidate-set modeling as a practical design principle for multimodal decision systems. Code is publicly available at `https://github.com/jin-s13/OpenVisualJev`.

## 1 Introduction

Vision–language models provide a general-purpose foundation for interpreting visual observations through natural-language instructions. Their integration into decision pipelines introduces a distinct inference requirement: converting this semantic understanding into low-latency, machine-consumable judgments. In this setting, the application specifies a decision space at runtime, and the model assigns probabilities to its alternatives. The emerging System One paradigm motivates direct prediction for such judgments, with visual decision models extending this interface to image-conditioned tasks (Yu & Yao, 2026; Zhou et al., 2026). This formulation brings the semantic flexibility of pretrained vision–language models to a structured probabilistic interface.

A central architectural question is how the runtime candidate set should enter the scoring function. Its elements have semantic content, but their presentation order is usually incidental to the decision. At the same time, the appropriate comparison can depend on which alternatives are available. A scorer should therefore preserve candidate identity under reordering while allowing relative preferences to change with the set. These requirements distinguish presentation symmetry from semantic interaction. Independent candidate scores provide the former, while joint comparison supplies the capacity needed for set-dependent decisions. Set representation learning offers a principled foundation for combining these properties (Zaheer et al., 2017; Lee et al., 2019).

We present OpenVisualJev[1], a multimodal decision model built on a vision–language backbone with a dynamic candidate-scoring head. The architecture separates visual grounding

[1] An independently developed research implementation of a Jev-style multimodal decision model.

of individual candidates from comparison within the current set. Tree-masked attention packs shared image and question prefixes with isolated candidate suffixes into one backbone forward. A position-free set module compares the resulting candidate representations, and a shared pointer maps them to a probability distribution over the supplied choices. This composition makes the scoring function permutation equivariant while retaining set-dependent preferences. Calibration-aware training and held-out temperature scaling complete the probability estimation pipeline. Figure 1 presents the model.

We evaluate order stability and set-dependent preferences through controlled tests, then assess standard recognition and VQA capability on the same Qwen3-VL backbone. The results connect candidate-set structure to stable decisions, stronger recognition with larger difficult candidate sets, and calibrated probability outputs.

Our contributions are threefold:

- We develop a multimodal decision model that combines pretrained visual–language representations with dynamic candidate scoring and an explicit probability-calibration pipeline.
- We formulate a candidate-scoring architecture that combines permutation-equivariant set interaction with shared pointer readout, and analyze its order stability and capacity for set-dependent preferences.
- We evaluate these properties through controlled structural tests, unified fine-grained recognition, and natural VQA, with ablations that characterize candidate interaction, candidate-count training, and probability calibration.

## 2 Decision interface and model structure

For an image $x$, a question $q$, and runtime-supplied candidate texts $C = (c_i)_{i=1}^{K}$, the model returns a normalized distribution over the current candidate slots. External question and candidate identifiers associate outputs with inputs; pointer prompts contain the image, question, and candidate text. This work studies mutually exclusive visual Choice decisions. The output distribution is defined over the supplied candidate set, and a held-out calibration stage adjusts its confidence.

### 2.1 Tree-masked multimodal encoding

We organize the input as a tree with a shared image prefix, a question prefix for each decision, and separate query and candidate suffixes. Common prefixes appear once in the packed sequence. A tree attention mask gives each token access to its causal predecessors within its segment and to all tokens on its ancestor path. Sibling candidate suffixes and distinct question branches are isolated. This organization shares visual and question context while preserving candidate-specific representations.

**Branch representations.** Let $E_\theta$ denote the image–text backbone and its final valid-token representation along a root-to-leaf path. The query leaf yields $u = E_\theta(x, q)$, and candidate leaf $i$ yields $h_i = E_\theta(x, q \parallel \texttt{Candidate:} \parallel c_i)$. These expressions identify the context visible to each leaf. A packed backbone forward computes all leaf representations using shared parameters. Visual features are injected at the shared image prefix. Each token retains its original branch-local multimodal position, so moving a candidate within the packed input preserves its positional coordinates and visible context.

### 2.2 Set interaction and dynamic pointer

The leaf representations ground each candidate in the image and question. A question-local set module then compares the available alternatives and produces their decision scores.

**Position-free candidate interaction.** Writing the candidate representations as rows of $H$, we apply a shared input projection followed by a position-free candidate module

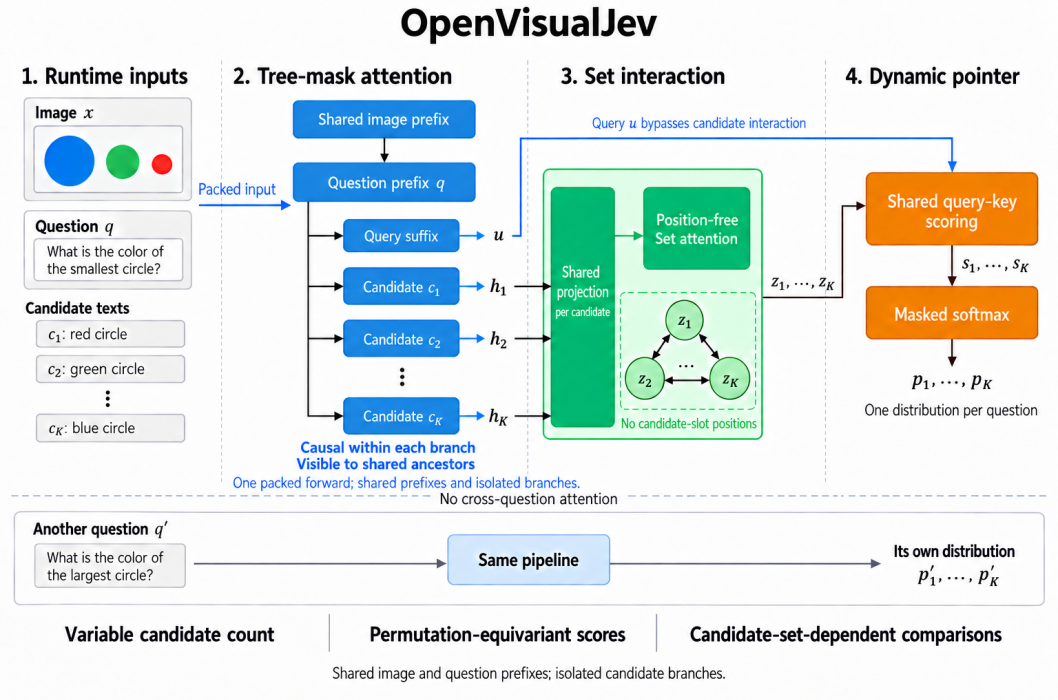



Figure 1: OpenVisualJev with tree-masked encoding. A packed input shares the image prefix and each question prefix across isolated query and candidate suffixes. Each token attends to its causal predecessors within its branch and to its ancestors. The shared projection is applied independently to each candidate representation; position-free Set attention then compares candidates jointly. The question query bypasses candidate interaction and scores the resulting keys before masked softmax. Each question has its own interaction and normalization. Training includes dataset-wise mean-confidence regularization, followed by held-out temperature scaling.

$S_\phi$. The Set module uses self-attention over candidate representations, followed by shared per-row transformations. The query representation remains separate from this candidate self-attention stack. Candidate logits and probabilities are

$$Z = S_\phi(H), \qquad s_i = \frac{\langle W_q \,\mathrm{LN}(u), W_k \,\mathrm{LN}(Z_i)\rangle}{\sqrt{d}}, \qquad p_i = \frac{\exp(s_i)}{\sum_{j\in\mathcal{V}} \exp(s_j)}, \tag{1}$$

where $d$ is the query–key dimension and $\mathcal{V}$ contains valid slots. Invalid padded keys are masked during attention and their output logits are excluded from normalization.

### 2.3 Structural properties

**Permutation-equivariant scoring.** In exact arithmetic, shared row-wise transformations commute with a candidate permutation matrix $P$. Self-attention obeys $\mathrm{Attn}(PH) = P\,\mathrm{Attn}(H)$ because query, key, and value rows, together with the key mask, are permuted consistently. Residual connections and row-wise normalization preserve this property. Consequently $s(PH) = Ps(H)$ and $p(PH) = Pp(H)$ when branch preprocessing and positions are unchanged and no candidate index is injected.

With a unique maximum, the selected candidate identity is preserved under candidate reordering.

**Question isolation and candidate-set dependence.** The tree mask isolates distinct question branches, while Set attention and probability normalization operate within each question. These two levels of isolation keep each question's scores independent of other

questions sharing the image prefix. Within each question, Set attention conditions every candidate representation on the full candidate set. The relative probability $p_i/p_j = \exp(s_i - s_j)$ therefore depends on set-conditioned logits. Changing the candidate set can change the relative preference between two existing candidates while preserving isolation from other questions.

### 2.4 Training and probability calibration

Cross-entropy supervises the correct candidate on each example. We complement this supervision with a group-level confidence objective during training and scalar temperature fitting after training.

**Calibration-aware training.** We train the scorer with cross-entropy and a dataset-wise mean-confidence regularizer, adapting the confidence–accuracy alignment principle of DCA (Liang et al., 2020). For an effective batch partitioned into dataset groups $B(d)$, let $a_n = \arg\max_i p_{n,i}$, $c_n = \max_i p_{n,i}$, and $r_n = \mathrm{I}[a_n = y_n]$. Let $\mathcal{D}$ denote the datasets represented in the effective batch. We optimize

$$\begin{aligned} \mathcal{L}_{\text{train}} &= \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{conf}}, \\ \mathcal{L}_{\text{conf}} &= \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \left[ \frac{1}{|B(d)|} \sum_{n \in B(d)} (c_n - \text{stopgrad}(r_n)) \right]^2 . \end{aligned} \tag{2}$$

Each group contributes equally. Balanced effective batches provide observations from every dataset, and the penalty aligns each group's mean confidence with its observed accuracy. Gradients flow through the selected probability; correctness supplies a detached target. The cross-entropy term retains supervision for individual candidate decisions. The training-loss ablation uses the same sampling and optimization settings with the penalty coefficient set to zero.

**Reserved probability calibration.** After fitting the scorer, we freeze its weights and fit a positive scalar temperature $T$ on a separate calibration split by minimizing candidate-event NLL. Deployment uses $p_i^{(T)} = \exp(s_i/T)/\sum_{j \in \mathcal{V}} \exp(s_j/T)$. A shared scalar adjusts confidence while preserving score ordering, predicted candidate identity, and permutation equivariance. Temperature fitting uses the held-out task and candidate-count distribution.

## 3 Experimental setup

All comparisons use Qwen3-VL-4B-Instruct (Bai et al., 2025). Vision parameters are frozen; language attention projections receive LoRA adaptation (Hu et al., 2021). We compare Set Pointer with a complete-candidate LM that reads all labeled alternatives and directly scores their answer tokens. Both directly output candidate probabilities. The main interface comparison combines candidate-conditioned cross-entropy and the dataset-wise calibration loss, optimized with AdamW (Loshchilov & Hutter, 2017). Independent Pointer is included in the relation test to probe the limitation of candidate-independent utilities. It uses independent candidate branches and a shared pointer readout. Section 4.1 details the input organization and scoring components of each interface.

The reported accuracy and probability tables were evaluated with ordinary batched branches using the same branch visibility and positional coordinates described in the method.

**Unified recognition.** We train one checkpoint per method across Pets (Parkhi et al., 2012), CUB, Cars (Krause et al., 2013), Dogs, Flowers (Nilsback & Zisserman, 2008), and Aircraft (Maji et al., 2013). The cleaned training, development, and calibration inventories contain 31,088, 3,637, and 3,635 images. The standard test uses 35,566 FOCI-style four-choice records (Geigle et al., 2024) distributed with Finedefics (He et al., 2025), with identical shuffled candidates for all methods. We retain the author's repeated Dogs test

row. Every question includes the gold class and a specified set of distractors, defining conditional multiple-choice classification. We reuse the published evaluation records and train both interfaces with our unified Qwen3-VL protocol.

**Main model and component ablation.** The main recognition and probability comparisons in Tables 3 and 6 use balanced hard-four-choice training. Each effective batch contains 96 questions, with 16 from each dataset. Both interfaces use seed 20261021 and the fixed epoch-three checkpoint after 2,016 updates, with identical candidate exposures. The full method uses calibration-loss weight five and a separate temperature minimizing equal-dataset calibration NLL. Table 7 isolates the training penalty and temperature within Set under this same recipe. Appendix A.1 specifies the optimization details.

**Natural VQA.** A separate supervised A-OKVQA experiment (Schwenk et al., 2022) evaluates ordinary task capability on an internal holdout of 2,576 questions, with disjoint training, selection, and calibration roles. We report the native LM comparison across three training seeds. Both interfaces are adapted separately to A-OKVQA. Split and optimization details are in Appendix A.

**Metrics.** We report accuracy, NLL, sum-Brier, and ECE. Each model's temperature is fitted only on reserved calibration data (Guo et al., 2017). Recognition averages datasets equally. Paired intervals resample image clusters within datasets; relation intervals resample scenes. Appendix A specifies resampling and calibration conventions. Bold values denote the best point estimates in the indicated comparison, including displayed ties.

## 4 Results and analysis

### 4.1 Order stability

We first test whether the scoring interface preserves candidate identity under changes in presentation order. We compare three scoring interfaces, summarized in Table 1. Their inputs, candidate interactions, and readouts are defined as follows.

**Independent Pointer** removes the candidate-interaction blocks and scores each independently encoded candidate using a shared 256-dimensional query–key pointer. **Set Pointer** adds two position-free Transformer blocks (four heads, fourfold feed-forward expansion, GELU, zero dropout), with the query bypassing candidate interaction. **Complete-candidate LM** encodes all labeled alternatives together and normalizes next-token logits over their single-token labels. All interfaces directly output candidate probabilities; Table 1 summarizes their structural properties.

| Interface | Candidate interaction | Equivariant scores |
|---|---|---|
| Independent Pointer | No | Yes |
| Set Pointer | Yes | Yes |
| Complete-candidate LM | Yes | Not guaranteed |

Table 1: Structural properties under the preprocessing and masking conditions in the method. Equivariance is a score property; ties and floating-point effects require separate decision-level checks.

For the complete-method checkpoints used in Table 3, LM changes its selected candidate on 108 of 755 questions (14.30%) across exhaustive permutations; Set changes on none. Each question is evaluated under all 24 candidate orders, yielding 18,120 predictions per model. Predictions are aligned by candidate identity before comparison. Figure 2 connects this order stability to the complete-method performance under larger difficult candidate sets.

### 4.2 Candidate interaction enables preference reversals

Paired three-choice questions share two objects but replace the third with a smaller or larger extreme, reversing the correct median. Candidate-independent utilities cannot reverse the shared objects' ranking. We test fixed checkpoints on 512 fresh scenes (2,048 questions), averaging three training seeds; a largest-object task controls for ordinary ranking.

| | Median relation | | | Maximum control | |
|---|---|---|---|---|---|
| Model | Acc. (%) | Pairs (%) | NLL | Acc. (%) | NLL |
| Independent | 50.0000 | 0.0000 | 0.70072 | 99.7721 | 0.01175 |
| Set | 99.0234 | 98.3073 | 0.25971 | 99.5768 | 0.03669 |
| Complete LM | **99.7721** | **99.5443** | **0.01878** | **99.8372** | **0.01062** |

Table 2: Fresh-scene test of fixed 80-epoch checkpoints, equally averaging the three retained training seeds. Pairs require both questions correct; all NLL values are uncalibrated. Set and complete-candidate LM both learn the median relation, which requires candidate interaction. The training budget is matched by optimizer updates.

Set learns the candidate-dependent median relation: its paired accuracy exceeds Independent by 98.3073 percentage points (whole-scene interval $[97.4609, 99.0234]$). Independent succeeds on the maximum control, isolating candidate interaction as the relevant distinction.

### 4.3 Standard recognition and candidate-count transfer

Table 3 evaluates the complete models after matched balanced six-dataset training and held-out calibration. Set reaches 90.35% mean accuracy versus 90.09% for LM, a +0.2690-point difference (paired interval $[+0.0086, +0.5317]$).

| Dataset | LM (%) | Set (%) | Difference (pp) |
|---|---|---|---|
| Pets | **95.88** | 95.42 | -0.46 |
| CUB | 84.52 | **85.57** | +1.05 |
| Cars | 93.74 | **93.83** | +0.09 |
| Dogs | 85.65 | **85.80** | +0.15 |
| Flowers | **99.11** | 98.99 | -0.11 |
| Aircraft | 81.61 | **82.51** | +0.90 |
| Equal-dataset mean | 90.09 | **90.35** | +0.27 |

Table 3: Unified mixed-dataset models on author-defined four-choice test records, using the same shuffled candidate sets for both methods. Each method uses one balanced CE-plus-calibration checkpoint across all six datasets, followed by its own held-out temperature. One training seed; differences are Set minus LM, computed before rounding.

**Transfer to larger difficult candidate sets.** Using the same checkpoints, we increase candidate counts with distractors selected by a fixed image–text ranking (Table 4). The Set–LM gap grows from 0.44 points at four choices to 1.80 at twenty (paired intervals $[0.17, 0.70]$ and $[1.42, 2.20]$). Larger difficult sets expose the benefit of joint comparison.

### 4.4 Training with variable candidate counts

We compare fixed four-choice training with counts sampled uniformly from $K \in \{2, 4, 10\}$, holding image exposure, updates, development selection, and test candidates fixed. Candidate-token computation varies with the sampled count; twenty-choice tests exceed the training range. Variable-count training improves twenty-choice accuracy by 2.19 percentage points for LM and 1.35 for Set. Set retains a 1.62-point advantage (paired interval

| Candidates | LM (%) | Set (%) | Difference (pp) |
|---|---|---|---|
| 2 | 93.47 | **93.85** | +0.38 |
| 4 | 90.10 | **90.54** | +0.44 |
| 10 | 84.62 | **86.00** | +1.38 |
| 20 | 80.26 | **82.06** | +1.80 |

Table 4: Difficult-candidate count transfer of the complete-method checkpoints in Table 3. Training uses four candidates; test candidates are mined with the fixed image–text ranking. Values are six-dataset mean accuracies from one shared training seed. Differences are Set minus LM.

| Test | LM fixed | Set fixed | LM variable | Set variable |
|---|---|---|---|---|
| Standard | 89.08 | 89.79 | 89.08 | **89.82** |
| Hard 2 | 92.87 | **93.36** | 92.73 | 93.26 |
| Hard 4 | 89.25 | **89.93** | 89.13 | 89.91 |
| Hard 10 | 84.94 | 86.27 | 86.02 | **87.24** |
| Hard 20 | 82.01 | 84.47 | 84.20 | **85.82** |

Table 5: Equal-dataset accuracy (%) for fixed four-candidate training and variable-candidate training. All four models use the same single training seed and test records. Difficult-distractor tests and the author-defined standard test are separate constructions.

$[1.31, 1.93]$), while standard accuracy changes little. The variable-count Set also preserves every decision in the exhaustive-order test.

### 4.5 Training and post-training calibration

Temperature scaling improves NLL and Brier for both models without changing accuracy (Table 6). After scaling, LM has lower NLL and Set has a lower Brier point estimate; the paired Brier difference interval includes zero. Per-dataset calibration curves appear in Appendix Figure 3.

| | Raw | | Temperature-scaled | |
|---|---|---|---|---|
| Metric ↓ | LM | Set | LM | Set |
| NLL | 0.3008 | 0.3478 | **0.2778** | 0.2831 |
| Sum-Brier | 0.1473 | 0.1480 | 0.1438 | **0.1428** |
| ECE | 0.0337 | 0.0448 | **0.0224** | 0.0355 |

Table 6: Equal-dataset mean probability metrics for the balanced CE-plus-calibration checkpoints in Table 3. ECE uses fifteen equal-count bins and is descriptive. Lower is better; bold marks the lowest value in each row.

**Training-time calibration.** Under the same balanced recipe, the training penalty improves raw NLL, Brier, and ECE (Table 7). Combining it with temperature scaling further reduces NLL and Brier relative to temperature-scaled CE training. The NLL difference is $-0.01539$ (paired interval $[-0.01953, -0.01142]$), with a small accuracy change of $-0.1273$ points. Appendix A.1 gives the recipe.

### 4.6 Natural VQA performance

On A-OKVQA, Set remains close to the native LM, with an accuracy decrease of 0.53 points across three seeds (Table 8; exploratory interval $[-1.7041, 0.6406]$). Scaled probability losses favor LM. This test evaluates retention of general VQA capability; split and uncertainty details are in Appendix A.

| Training | Probability | Accuracy (%) | NLL ↓ | Brier ↓ | ECE ↓ |
|---|---|---|---|---|---|
| CE | Raw | **90.48** | 0.4003 | 0.1519 | 0.0521 |
| CE + calibration | Raw | 90.35 | 0.3478 | 0.1480 | 0.0448 |
| CE | Temperature | **90.48** | 0.2985 | 0.1457 | 0.0427 |
| CE + calibration | Temperature | 90.35 | **0.2831** | **0.1428** | **0.0355** |

Table 7: Training-loss comparison for Set under balanced six-dataset training with one shared seed. Metrics are equal-dataset means on the standard test; Brier is summed over candidates and ECE uses fifteen equal-count bins. Each model receives its own held-out temperature fitted with an equal-dataset NLL objective. Bold marks the best value in each metric column, including ties.

| Interface | Accuracy (%) | Scaled NLL | Scaled Brier |
|---|---|---|---|
| Native LM | **87.2800** | **0.3703** | **0.1845** |
| Set | 86.7495 | 0.3819 | 0.1932 |

Table 8: Natural VQA capability and calibrated probability quality, using the completed supervised holdout experiment. Means use all registered seeds and frozen, independently fitted temperatures. The comparison follows the registered native-LM secondary analysis. Raw probability differences and protocol details appear in Appendix A.

## 5 Related work

**Multimodal decision models.** Recent Jev-style systems map runtime-defined alternatives to probabilities. Visual Jev and PixelJev adapt vision–language models using option-token readouts (Yu & Yao, 2026; Zhou et al., 2026), while Koa-action learns atomic label tokens with randomized label mappings (Dai et al., 2026). OmniJev combines learned option scoring with probabilistic supervision and calibration (Xu et al., 2026), and Chinese-Jev develops a text-encoder decision model with heterogeneous supervision (Wang et al., 2026). Shared-context execution provides a complementary systems direction: Visual Jev reuses image prefixes, and FastJEV studies state reuse and candidate-layer pruning (Yu & Yao, 2026; Ma et al., 2026). OpenVisualJev focuses on the scoring architecture, combining independent candidate representations with explicit, permutation-equivariant joint comparison.

**Set-dependent choice.** Deep Sets establishes invariant and equivariant set representations, and Set Transformer models interactions through attention (Zaheer et al., 2017; Lee et al., 2019). Context-dependent choice models such as FETA and FATE allow an alternative's utility to depend on the surrounding set, including preference reversals when another alternative changes (Pfannschmidt et al., 2019). Our candidate module applies these principles to multimodal representations. The relation experiments test this capacity directly, alongside the stability of scores under candidate permutations.

**Probability quality and decision evaluation.** Jev benchmarks and decision-gate studies examine calibration, selective prediction, option ordering, and candidate-count robustness (Deußer et al., 2026; Rafe & Das, 2026). Judge routing, security screening, and mobile action selection illustrate the downstream use of decision scores (Li et al., 2026; Liu, 2026; Zhang, 2026). Visual MCQA debiasing and selective VQA likewise connect answer confidence with reliable selection (Atabuzzaman et al., 2025; Eisenschlos et al., 2024). Our evaluation separates recognition accuracy, order stability, and probability quality. The calibration pipeline builds on confidence–accuracy regularization and multiclass calibration objectives (Liang et al., 2020; Hebbalaguppe et al., 2022), followed by temperature scaling (Guo et al., 2017); ablations measure the contributions of training-time and post-training adjustment.

**Fine-grained visual recognition.** FOCI formulates fine-grained recognition as multiple-choice answering with difficult distractors (Geigle et al., 2024). Finedefics combines

attribute-mediated alignment with classification instruction tuning (He et al., 2025). We reuse the candidate records released with the Finedefics evaluation code and train a unified Qwen3-VL model across six datasets. This provides a common recognition setting for comparing scoring architectures and complements the controlled structural experiments.

## 6 Conclusion

OpenVisualJev combines tree-masked multimodal encoding, permutation-equivariant set interaction, and calibrated dynamic pointer scoring. Controlled experiments establish decision stability and candidate-dependent comparison, while recognition and VQA tests assess standard task capability. Variable-count training and probability calibration further improve robustness and decision scores. These results support explicit candidate-set modeling as a practical design for multimodal decision systems.

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## AI usage disclosure

Generative AI assisted this research under the author's direction, using the open-source `ai-research-writing-skill` workflow. Assistance covered literature review, method and experiment design, implementation and debugging, analysis, figures, and manuscript preparation. Synthetic relation data were produced by executable generation code; reported results were computed from model predictions and checked against evaluation artifacts. The author reviewed the work and retains responsibility for its claims, code, data, and figures.

## A Essential reproducibility details

**Architecture and scoring.** The backbone hidden width is 2560. Candidate summaries are projected to width 256; Set uses two pre-normalized position-free Transformer blocks, four attention heads, feed-forward expansion four, GELU, and zero dropout. Independent removes cross-candidate blocks. The separate question query bypasses candidate attention. The readout operates in single precision over bfloat16 backbone features. The LM scores the next-token logits of single-token candidate symbols, directly normalized over the supplied alternatives. Pointer branches omit candidate indices. Pointer uses separate candidate contexts, while LM uses one complete-candidate context.

**Recognition data and optimization.** Pets, CUB, Cars, and Dogs reserve development/calibration images within classes from the official training-side inventory. Flowers and Aircraft use official validation images for those roles. Exact and reviewed near-duplicate groups are separated from test-side images; additional cross-dataset exclusions precede mixed training. The duplicate audit covers the task-specific training and evaluation inventories. The six datasets contain 37, 200, 196, 120, 102, and 100 classes, respectively; the choice count is independent of class count.

FOCI/Finedefics test candidate sets are preserved, including the original repeated Dogs row, and shuffled consistently across methods. The unified protocol follows the released Dogs test inventory. Training and dynamic tests mine difficult incorrect labels with the recorded OpenCLIP ViT-L-14 image–text ranking; random distractors are distinct sampled incorrect classes. The correct label is included. The standard author-defined four-choice test and the newly constructed difficult four-choice test are separate conditions.

Unified training uses learning rate 0.0001, accumulation over eight questions, three epochs, cosine decay, warmup fraction 0.03, AdamW weight decay 0.01, global clipping at norm one, and language-attention LoRA rank/scaling 16/32. The seed is 20261021. Both methods use candidate-conditioned cross-entropy, the same exposure and selection rule, and development-based checkpoint selection. Variable-count training changes only the registered candidate-count sampling and associated candidate construction; candidate-token computation follows the sampled candidate counts. Images use processor area bounds 65536 and 262144 pixels; Aircraft crops the bottom twenty pixels. Other recognition images are uncropped.

**Relation construction and endpoint.** Images contain colored objects with distinct sizes. Paired three-choice questions share two middle objects but replace the third candidate with a smaller or larger extreme, reversing the median answer. The largest-object task serves as an independent-ranking control. Training updates language LoRA and the trainable pointer head using choice-label supervision. The original training inventory has 256 scenes; the final test uses fresh scenes to evaluate the fixed training endpoint. Seeds are 20261011, 20261012, and 20261013. The common fixed endpoint is 80 epochs, using AdamW learning rate 0.0003, accumulation over 64 questions, weight decay 0.01, and clipping at norm one. This longer-budget test followed earlier development diagnostics; it evaluates the final achievable capability under this recipe. All final seeds are retained. Whole-scene bootstrap intervals condition on those trained seeds.

**VQA splits and training.** A-OKVQA training uses 12769 questions over 12404 images; selection uses 859 questions over 827 images, and calibration uses 852 questions over 827 images. The internal holdout has 2576 questions over 2482 images, grouped into 2481 resampling components after a reviewed near-duplicate merge. It is split from the official training inventory. Seeds are 20260924, 20260927, and 20260930. Models train for three epochs with learning rate 0.0002 and accumulation over eight questions; LoRA, optimizer, warmup, and clipping settings otherwise follow the shared recipe above. Checkpoint selection maximizes answer-text correctness, then minimizes NLL, then prefers the earlier epoch. Blank and duplicate choices are retained; probability mass for duplicate literal answer strings is summed when computing answer-event losses. Images are uncropped.

**VQA comparison scope.** The main text presents the registered native-LM secondary comparison. The original primary contrast used a canonically sorted LM input: Set-minus-canonical-LM accuracy was $-0.5694$ percentage points, with conditional interval $[-1.2861, 0.1289]$. The original primary and secondary analysis roles are retained. For native LM, raw Set-minus-LM NLL and Brier differences are $+0.1933$ and $+0.0243$; their exploratory crossed intervals are $[0.0779, 0.3260]$ and $[0.0078, 0.0407]$. Scaled differences are $+0.0116$ for NLL and $+0.0087$ for Brier, with crossed intervals $[-0.0054, 0.0284]$ and $[-0.0025, 0.0204]$. The registered analysis estimates accuracy and probability-loss differences with intervals.

**Calibration and statistical conventions.** Recognition minimizes reserved-set pooled candidate NLL over $\log T \in [-5, 5]$ using bounded scalar optimization and explicit boundary checks. VQA uses a fixed grid of 2001 log-temperatures in the same range, preferring the value closest to zero on objective ties. Temperatures are fixed before test scoring. Sum-Brier sums squared probability errors across candidate or literal-answer events. Recognition ECE uses fifteen equal-count bins; VQA uses fifteen equal-width bins. Exact score ties are resolved by selecting the first slot. Paired bootstrap samples keep repeated images and related questions together. The recognition and relation intervals condition on the trained seeds; the crossed VQA analysis is exploratory with few training realizations. All intervals are conditional on the fitted temperatures and the evaluation distribution.

**Reproduction scope.** The recorded implementation uses PyTorch 2.8.0, Transformers 4.57.3, PEFT 0.17.1, and NVIDIA A800 devices. Data manifests, model checkpoints, raw predictions, and analysis scripts provide the evidence trail. The project archive preserves earlier specialist adapters, intermediate training endpoints, optimization probes, and engineering validation records. Public external reproduction remains a planned release milestone.

### A.1 Training-time calibration protocol

The loss comparison uses balanced six-dataset hard four-choice training, matched candidate exposures, and a shared seed of 20261021. Each effective batch contains 96 questions, with 16 from each dataset. Both arms train for three epochs (2,016 updates and 193,536 question exposures) and use the final epoch-three checkpoint. The learning rate is 0.0001, with 61 warmup updates; the remaining optimizer and LoRA settings follow the recognition recipe.

The coefficient of five is the evaluated development setting. The balanced recipe was developed after inspecting earlier benchmark results. Cross-entropy is augmented with a weight of five times the average dataset-wise squared gap between mean top-choice probability and empirical accuracy in the effective batch; the correctness target is detached. The control sets this weight to zero. Each model is evaluated on the same reserved calibration and standard test splits. Its positive scalar temperature minimizes equal-dataset mean calibration NLL. Paired image-cluster bootstrap intervals condition on these trained weights and fitted temperatures. Tables 3 and 6 compare the complete LM and Set interfaces under this same balanced recipe.

### A.2 Equivalent answer events

For the registered VQA protocol, blank and duplicate candidate strings are preserved. Prediction selects the maximum-logit slot, with first-slot tie breaking, before comparison by literal answer text. For probability scoring, the mass of identical strings is summed. Given a gold answer event $G$, the candidate-conditioned loss is

$$\mathcal{L} = -\log \sum_{i \in G} p_i = \mathrm{LSE}_{j \in \mathcal{V}}(s_j) - \mathrm{LSE}_{i \in G}(s_i). \tag{3}$$

Labels supply supervision to the loss and evaluator; forward prompts contain the image, question, and candidate text. This convention preserves the official answer-text task while keeping slot and answer-event probabilities distinct.

### A.3 Additional visual analyses

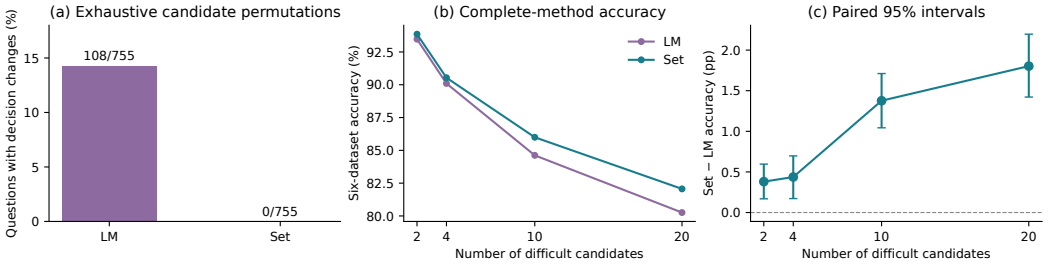



Figure 2: Complete-method structural analysis. (a) Fraction of questions whose predicted candidate identity changes across all twenty-four orders. (b) Six-dataset mean accuracy on difficult candidate sets. (c) Set-minus-LM accuracy with marginal paired image-cluster bootstrap intervals. All panels use the same complete-method checkpoints as Table 3; the training candidate count is four.

### A.4 Supplementary evaluation details

**Candidate-count study.** The variable-count training ablation uses a separate unified CE recipe: three epochs with matched image exposure, alternating difficult and random distractors, and development selection by equal-dataset accuracy with earlier-epoch tie-breaking. Fixed-count training uses four candidates; variable-count training samples uniformly from $K \in \{2, 4, 10\}$. The reported count-transfer tests use difficult distractors at two, four, ten, and twenty candidates alongside the standard records. The four checkpoints share the training seed and test manifests. This study matches image exposure and updates; candidate-token computation varies with candidate count. Twenty-candidate testing measures transfer beyond the training count range.

**Order and relation tests.** The order test evaluates all 24 permutations of four candidates on 755 class-stratified questions, aligning candidate identities before comparison. Both this test and the difficult-candidate count-transfer evaluation use the complete-method checkpoints from Table 3, with unchanged weights and inference precision. Relation tests use

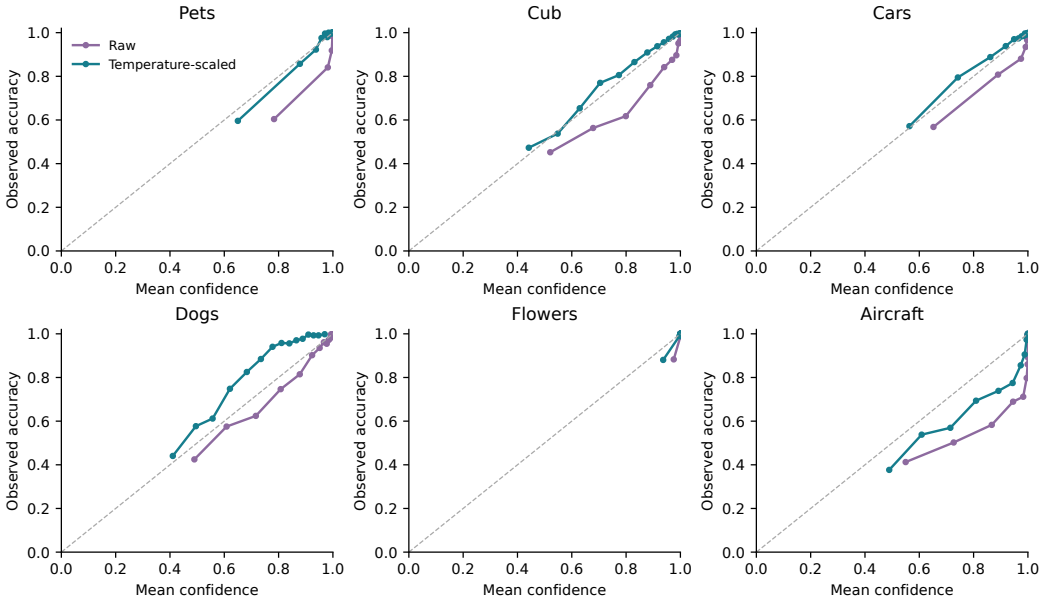



Figure 3: Probability calibration of the complete Set model on the six recognition datasets. Each curve uses fifteen equal-count confidence bins; points show mean top-choice confidence and observed accuracy. The dashed diagonal denotes matched confidence and accuracy. Temperature scaling uses the single scalar fitted on the reserved calibration data, with unchanged predictions.

synthetic scenes with paired candidate sets that reverse the required preference between shared objects. Independent, Set, and complete-candidate LM are evaluated with three training seeds at the same fixed final training endpoint on fresh scenes. The paired metric requires both questions correct; a largest-object control tests ordinary independent ranking.

**Probability quality and uncertainty.** Each model receives its own scalar temperature fitted only on reserved calibration data (Guo et al., 2017). We report accuracy, NLL, sum-Brier, and ECE, distinguishing raw and scaled probabilities. Recognition ECE uses fifteen equal-count bins; the VQA protocol uses equal-width bins over literal answer events. Recognition intervals use 10,000 paired image-cluster bootstrap replicates within datasets and equal dataset weighting; repeated image rows stay together. They condition on the selected weights and fitted temperatures, excluding training-seed and calibration-fit uncertainty. Relation intervals resample whole scenes; the VQA interval additionally resamples the few available seeds and is exploratory. Intervals are marginal, without multiplicity correction. Further protocol details appear in Appendix A.